

PROJECT BASED LEARNING · SEMESTER 7



Linux Network Diagnostic Tool

NetDiag · A Bash-Based Network Health Monitoring Utility

Nirav Panchal · Information Technology · 231250116034

Shree Swaminarayan Institute of Technology (SSIT)
Affiliated to Gujarat Technological University (GTU)

One command replaces many manual checks

Connectivity faults can originate at the interface, default gateway, DNS, or Internet. Checking each manually needs multiple Linux commands.

User	NetDiag	Interface	Gateway	DNS	Internet	Local Ports	Overall Status
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NetDiag combines the checks into a single CLI utility for a quick network health overview.

Implemented entirely in Bash using standard Linux networking utilities.

Objectives



Detect the active network interface



Display IPv4 address and interface state



Detect and test the default gateway



Check DNS configuration and resolution



Test Internet connectivity and measure latency



Check common local TCP ports such as 22, 443 and 80



Generate an overall network health status



Maintain a diagnostic log for future reference

Tools and technologies used

OPERATING SYSTEM

Ubuntu / Linux

PROGRAMMING

Bash shell scripting

Linux utilities

ip · ping · ss
resolvectl · dig

Text processing

grep · awk · sed · cut

Environment

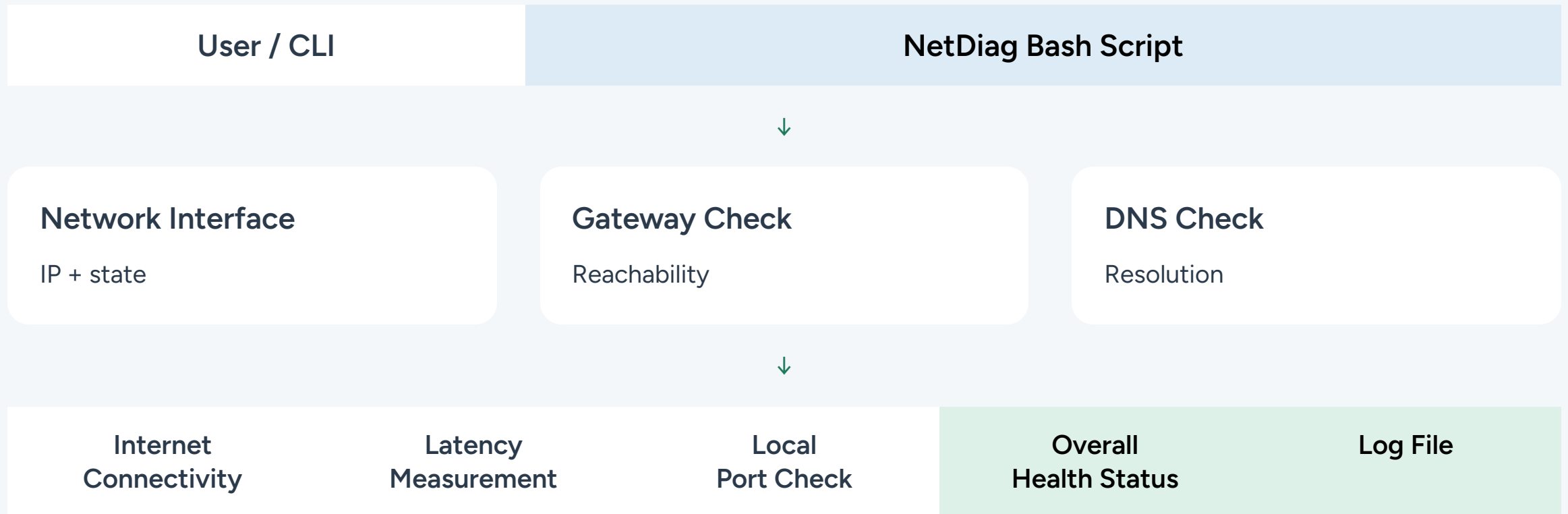
Linux terminal
Nano / text editor

Also

Log files
Command-line interface (CLI)

System design: from CLI to log file

NetDiag gathers data from Linux networking utilities, processes it with Bash commands, displays the diagnosis, and stores a summary in a log file.



Key parts of the Bash program

INTERFACE

```
INTERFACE=$(ip route | grep '^default' | awk '{print $5}')
```

GATEWAY

```
GATEWAY=$(ip route | grep '^default' | awk '{print $3}')
```

IP

```
IP_ADDRESS=$(ip -4 addr show "$INTERFACE" | grep 'inet ' | awk '{print $2}' | cut -d/ -f1)
```

INTERNET

```
PING_RESULT=$(ping -c 1 -W 2 "$INTERNET_TARGET" 2>/dev/null)
```

PORTS

```
ss -lnt
```

HEALTH

```
if [ "$STATE" = "UP" ] && [ "$GATEWAY_STATUS" = "REACHABLE" ] && [ "$DNS_STATUS" = "WORKING" ] && [ "$INTERNET_STATUS" = "ONLINE" ]
```

Bash combines Linux networking commands, conditional logic, pipes, text processing, and logging into a single diagnostic workflow.

Output screenshots

Terminal Output

NetDiag · Screenshot placeholder

Shows the interface, gateway, DNS, Internet, latency and port checks with the overall status.

```
nirav-panchal@Nirav-Ubuntu-Linux:~$ ./netdiag.sh

=====
      LINUX NETWORK DIAGNOSTIC
=====

[1] Network Interface
    Interface : wlp46s0
    State      : UP
    IP         : 192.168.1.5

[2] Default Gateway
    Gateway    : 192.168.1.1
    Status     : REACHABLE

[3] DNS
    DNS Server : 1.1.1.1
    Status     : WORKING

[4] Internet Connectivity
    Target     : 8.8.8.8
    Status     : ONLINE
    Latency    : 13.2 ms

[5] Local Port Check
    SSH (22)   : CLOSED
    HTTP (80)  : CLOSED
    HTTPS (443): CLOSED

=====
Overall Status : HEALTHY
=====
```

Help / Usage Output

Screenshot placeholder

Shows the usage and options text displayed by the help option.

```
nirav-panchal@Nirav-Ubuntu-Linux:~$ ./netdiag.sh --help
Linux Network Diagnostic Tool

Usage:
./netdiag.sh      Run network diagnostics
./netdiag.sh --help Show this help
```

Log File Output

Screenshot placeholder

Shows the diagnostic summary saved to the log file.

```
nirav-panchal@Nirav-Ubuntu-Linux:~$ cat logs/netdiag.log
2026-10-06 19:15:07
Interface : wlp46s0
IP        : 192.168.1.5
Gateway   : REACHABLE
DNS       : WORKING
Internet  : ONLINE
Latency   : 17.3 ms
Overall   : HEALTHY
-----
2026-10-07 05:47:42
Interface : wlp46s0
IP        : 192.168.1.5
Gateway   : REACHABLE
DNS       : WORKING
Internet  : ONLINE
Latency   : 13.2 ms
Overall   : HEALTHY
-----
```

Expected fields

Network Interface · Interface State · IPv4 Address · Default Gateway · Gateway Status · DNS Server · DNS Status · Internet Status · Latency · Local Port Status · Overall Status

How the diagnostic works

- 01 Identify the default network interface
- 02 Retrieve IPv4 address and interface state
- 03 Identify and ping the default gateway
- 04 Check DNS configuration and perform resolution
- 05 Ping an Internet target and calculate latency
- 06 Inspect local TCP listening ports
- 07 Determine overall network health
- 08 Save diagnostic information to the log file

Key features and results

Single-command network diagnosis

Fast identification of common connectivity problems

Gateway and DNS

Gateway reachability and DNS functionality testing

Internet

Internet connectivity and latency measurement

Local service / port visibility

Persistent diagnostic logging

Help option and dependency checking

Overall status

HEALTHY / WARNING

Demonstrates practical Linux administration and Bash scripting skills.

Conclusion

NetDiag simplifies basic Linux network troubleshooting

- Combines multiple standard networking utilities into one CLI tool
- Demonstrates Bash scripting, pipelines, conditional logic, networking concepts, and logging
- A practical foundation for advanced Linux administration and DevOps tooling

GitHub Repo:

niravp-06/**Linux-Network-Diagnostic-Tool**

PBL – Semester 7

1 Contributor

0 Issues

0 Stars

0 Forks



GitHub

GitHub – niravp-06/Linux-Network-Diagnostic-Tool: PBL – Semester 7

PBL – Semester 7. Contribute to niravp-06/Linux-Network-Diagnostic-Tool development by creating an account on GitHub.

Thank You

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